

MRM walkthrough on OTIS provincial data

What this vignette covers

The **MRM (McNamara–Ruhela–Medina)** framework is a coordinated set of ten causal estimators paired with a multi-source data layer for Canadian carceral, police, and oversight data. This vignette uses the provincial Ontario Tracking Information System (OTIS) restrictive-confinement microdata as an example, applies the ten-estimator ensemble to a binary-treatment design on dataset `a01`, and shows how to read the resulting summary.

The mathematical foundations are developed in the companion paper (Ruhela 2026, *The MRM Framework*, Zenodo 10.5281/zenodo.20096075).

Loading OTIS

OTIS is shipped with the package; the `morie_load_dataset()` loader hides the SQLite-backed indirection.

```
library(morie)
otis <- morie_load_dataset("otis-2025-a01")
str(otis)
```

The canonical a01 design

For dataset `a01` the canonical formulation is `T_high_ac` (a binary treatment derived from administrative-classification flags) on `Y_vm_count` (a count of a specific in-confinement observation) with the standard demographic covariate set. This is the design choice that the per-row MRM modules implement.

```
# Full ten-estimator ensemble on the canonical a01 design:
result <- estimate_ate(
  data      = otis,
  outcome   = "Y_vm_count",
  treatment = "T_high_ac",
  covariates = c("age", "sex", "region", "fiscal_year")
)
print(result)
```

The returned object summarises the IPW (Hájek), AIPW (Robins–Rotnitzky–Zhao), g-computation, propensity-score-matching (1:1 NN and five-strata subclass), IRM-DML (Chernozhukov *et al.* 2018), PLR-DML, and SuperLearner-stacked AIPW estimates. Multi-SE comparison (pooled, cluster on fiscal year, cluster on individual ID, two-way) is reported alongside the IRM-DML primary.

Augmented IPW

```
result_aipw <- estimate_aipw(  
  data      = otis,  
  outcome   = "Y_vm_count",  
  treatment = "T_high_ac",  
  covariates = c("age", "sex", "region", "fiscal_year")  
)  
print(result_aipw)
```

Aggregate companion: incidence-rate ratios

For aggregate (year-level) outcomes the analog is a Poisson or negative-binomial GLM with cluster-robust standard errors. The MRM framework reports both the per-row individual-level estimate (above) and the aggregate IRR family in parallel; see the companion paper for the formal aggregate-IRR notation.

Mandela classification

A separate Mandela-Rules classifier (UN Mandela Rules 43 and 44) is applied at both the federal and provincial levels. The provincial implementation uses a duration-only proxy that is documented explicitly in the framework paper. Federal counterpart analyses (SIU IAP, Sprott–Doob–Iftene) live in the companion Python module `morie.tps_csi` and `morie.siu_iap`.

Where to go next

- The full MRM framework paper, including all ten estimators, multi-SE comparison, propensity calibration, and the Sprott–Doob–Iftene replication tables, is at Zenodo [10.5281/zenodo.20096075](https://zenodo.org/record/105281/files/mrm_framework.pdf).
- The MORIE package paper covers the broader toolkit: [10.5281/zenodo.20096350](https://zenodo.org/record/105281/files/morie_package_paper.pdf).
- Citation: see `citation("morie")`.